BRIJESH KUMAR GAUR

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PROFESSIONAL SUMMARY

B.Tech Electronics & Communication Engineering student at NIT Srinagar (Class of 2028) with a foundation in C++, Java, Python, Data Structures & Algorithms, digital electronics, communication systems and embedded systems. Hands-on project experience spanning algorithmic systems, digital design, quantitative analysis and robotics. Active competitive programmer with interests in backend engineering, embedded systems, UAV/drone technology and core ECE.

EDUCATION

- **National Institute of Technology, Srinagar** — Aug 2024 – June 2028
- Bachelor of Technology — Electronics & Communication Engineering | **CGPA: 7.07** (through 4th semester)
- **St. Thomas Inter College** — 2021 – 2023
- Class 12 (UP Board) — **87.4%** | Class 10 (UP Board) — **85%**

TECHNICAL SKILLS

- **Languages:** C++, Java, C, Python, Verilog
- **Backend & Databases:** MySQL, DBMS, SQL, Spring Boot, REST API Design
- **Web Development:** HTML, CSS, JavaScript
- **Core CS:** Data Structures & Algorithms, Object-Oriented Design, Problem Solving
- **ECE / Embedded:** Digital Electronics, Microprocessors, Embedded Systems, UAV Systems
- **Tools:** Git, GitHub, VS Code, IntelliJ IDEA, CLion, Arduino

PROJECTS

Git Inspired Version Control System | C++, Data Structures & Algorithms | Prof. Rohit Vaish (2025)

- Engineered an in-memory C++ version control system, achieving O(1) complexity for text edits, commits, and rollbacks
- Architected n-ary trees for commit tracking and chaining hash maps for O(1) state lookups across 1M indices
- Integrated max heaps and string-hashing for real-time analytics, enabling O(log N) metric updates and file queries

Low Latency Limit Order Book Matching Engine | C++, Data Structures & Algorithms | Self Project (2026)

- Developed a C++ matching engine processing 4,000,000+ orders per second at a 246-nanosecond average latency
- Engineered custom memory structures using Hash Maps and Doubly Linked Lists, achieving O(1) order cancellations
- Integrated GNU PBDS Red-Black Trees to enforce price-time priority, optimizing order placements to O(log N)

SocialNet Simulator | C++, Data Structures & Algorithms | Prof. Rohit Vaish (2025)

- Built a C++ social network simulator, utilizing O(1) unordered maps to manage user profiles and connections
- Engineered custom AVL trees to store friend networks, achieving O(log N) insertions and alphabetical sorting
- Implemented self-balancing trees for post timelines, enabling O(log N) complexity for recent post retrievals

Synchronous Digital Counter Design | Verilog, Digital Electronics | Prof. Manan Suri and Prof. Anuj Dhawan (2026)

- Designed and implemented 4-bit synchronous Gray-code counter, defined initial states using SR and D flip-flops
- Calculated optimized flip-flop input logic using detailed state tables, K-maps, and Boolean minimization techniques
- Implemented the Verilog modules on a CPLD board, verifying the physical hardware against theoretical state tables

Monte Carlo Portfolio Optimizer | Python, Quantitative Finance | Self Project (2026)

- Built a portfolio optimizer using Monte Carlo simulations to maximize Sharpe ratio and aggressive returns for optimal allocation
- Calculated metrics including expected return, volatility, and Sharpe ratio using Modern Portfolio Theory and covariance math
- Visualized the efficient frontier to interpret portfolio performance and compare defensive vs. aggressive growth strategies

Mini CNC Plotter | Arduino, Embedded Systems and Robotics | Prof. Sagar Sarkar (2025)

- Designed and built a CNC plotter using Arduino UNO, stepper motors, and servo motors with the help of GRBL C++ Library
- Implemented G-code (XY control) and M-code (Z-axis control) execution with GRBL for converting them into machine instructions
- Created and optimized STL files, fabricated parts using 3D printing, and integrated hardware–software communication

ACHIEVEMENTS

- Participated in a college-level coding contest at NIT Srinagar
- Actively solving problems on Codeforces to strengthen DSA and competitive programming skills
- **2nd Position** — Constitution Day Essay Writing Competition

EXTRA-CURRICULAR ACTIVITIES

- Casual football and cricket player

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B.Tech Electronics & Communication Engineering | NIT Srinagar | Class of 2028

COURSES DONE — ON CAMPUS THROUGH 4TH SEMESTER

- Linear Algebra
- C Language
- Mathematics — 2nd Semester
- Mathematics — 3rd Semester
- Mathematics — 4th Semester
- Electronics
- Digital Electronics & Logic Design
- Network Analysis
- Applied EMFT & Waves
- Signals & Systems
- Communication Systems
- Digital Communication
- Control Systems
- EEM
- Philosophy for Engineers

COMPETITIVE PROGRAMMING / PROBLEM SOLVING

- Regular practice of Data Structures & Algorithms using C++ and Codeforces
- Practice includes binary search, arrays, matrices, subarrays, graph algorithms, backtracking and optimization techniques
- Focus on algorithmic complexity, optimized solutions and implementation accuracy

ECE / TECHNICAL INTERESTS

- Embedded Systems & Microcontrollers
- UAV / Drone Technology
- Backend Development
- Digital Design and Hardware Systems
- Core ECE / Semiconductor Research
- AI/ML and Data-Driven Engineering

ADDITIONAL INFORMATION

- Current academic status: 3rd year, B.Tech ECE, NIT Srinagar
- Targeting winter and summer internship opportunities in ECE, embedded, UAV, semiconductor and software roles